

Supporting Information for “Estimates of Lunar Regolith Thermal Conductivity K_d at the Apollo 15 and 17 Heat-Flow Boreholes”

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This supplement collects the held-out validation diagnostics and the Bayesian K_d – Q_b cross-check referenced in §3.3 of the main text. Every figure and number is regenerated by the public repository (<https://github.com/rp3gregorio/Lunar-HFE>); the generating script is named in each caption.

S1 Certification of the periodic-equilibrium solver

The retrieval forward model computes the periodic steady state of the flux-bottom-boundary column by the flux-anchored outer iteration described in §2.3 of the main text. Figure S1 certifies its two defining properties at the retrieved per-site K_d^* values. (a) *Initial-guess independence*: converged cycle-mean profiles started from 240 K and 260 K agree to ≤ 19 mK at every depth in the 5-m column — the initial thermal state leaves no measurable imprint on the retrieval. (b) *Fixed-point drift*: a 120-lunation verification integration started from the converged profile drifts by ≤ 73 mK at the sensor depths, bounding the residual solver systematic on K_d^* at $\lesssim 0.15$ mW m $^{-1}$ K $^{-1}$, an order of magnitude below the bootstrap uncertainty. Both tests are regenerated by `scripts/figures/make_equilibrium_certification.py`.

S2 Held-out validation of the per-site retrieval

Two internal checks confirm that the per-site K_d^* retrieval is not driven by a single sensor or sensor type. (i) *TG vs. TR cross-prediction*: fitting K_d on the gradient-bridge (TG) sensors alone and evaluating on the ring-bridge (TR) sensors, and vice versa. (ii) *Leave-one-deepest-out*: withholding the deepest sensor at each site ($z = 139$ cm at Apollo 15, $z = 234$ cm at Apollo 17), refitting K_d on the remainder, and predicting the withheld temperature. Results are summarised in Figure S2; the out-of-sample degradations are quoted in the main text (§3.3).

S3 Bayesian K_d – Q_b cross-check

The non-parametric bootstrap is the primary uncertainty estimate in the main text because it propagates only quantities measured at the boreholes. As an independent check on the *direction* of the inter-site contrast, we also sample the joint posterior of (K_d, Q_b) at each site with an affine-invariant MCMC ensemble sampler (Foreman-Mackey et al., 2013), placing the published basal-flux envelopes (Langseth et al., 1976; Saito et al., 2007; Nagihara et al., 2018) as priors on Q_b and using the deep-sensor Gaussian likelihood built from the cached $\text{RMSE}(K_d)$ curves. The posteriors trace the K_d/Q_b degeneracy ridge discussed in §4.2 of the main text; the marginal medians differ from the bootstrap point estimates by construction (the Q_b prior slides each marginal along the ridge), but the ordering $K_d^{\text{A17}} > K_d^{\text{A15}}$ is preserved with high posterior probability (value quoted in §3.3 of the main text).

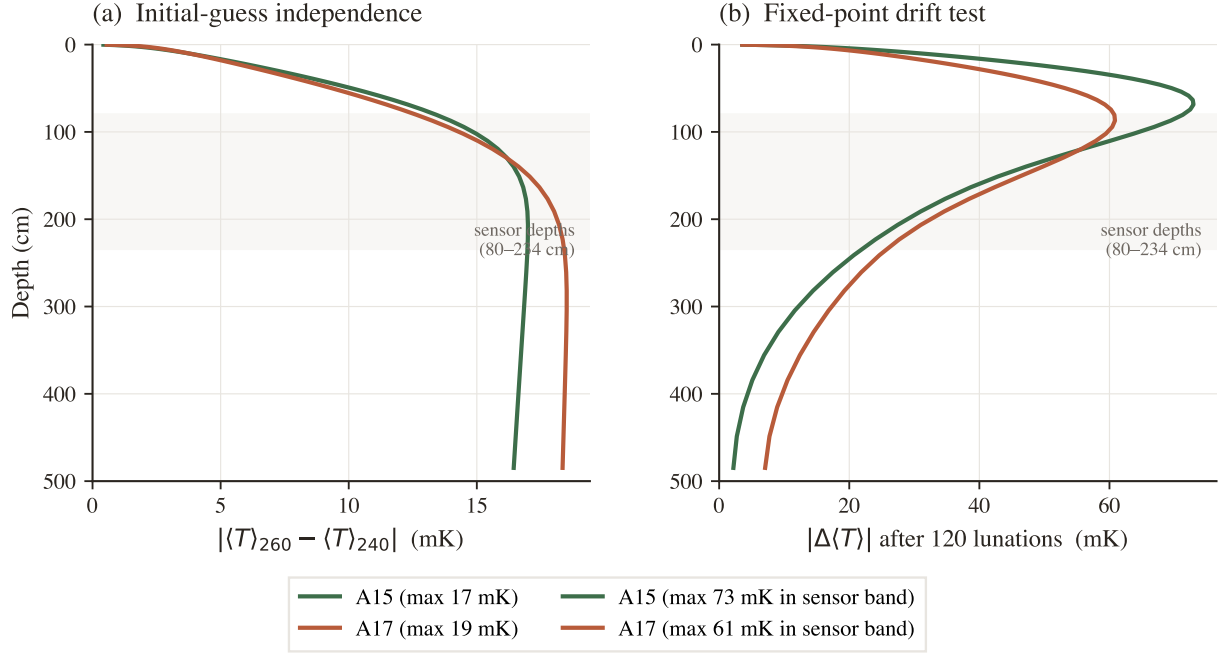


Figure S1. Solver certification at the retrieved K_d^* . (a) Absolute difference between converged cycle-mean profiles from initial guesses of 240 and 260 K. (b) Absolute change of the cycle-mean profile over a 120-lunation verification integration started from the converged state. Shaded band: HFE sensor depths used in the retrieval.

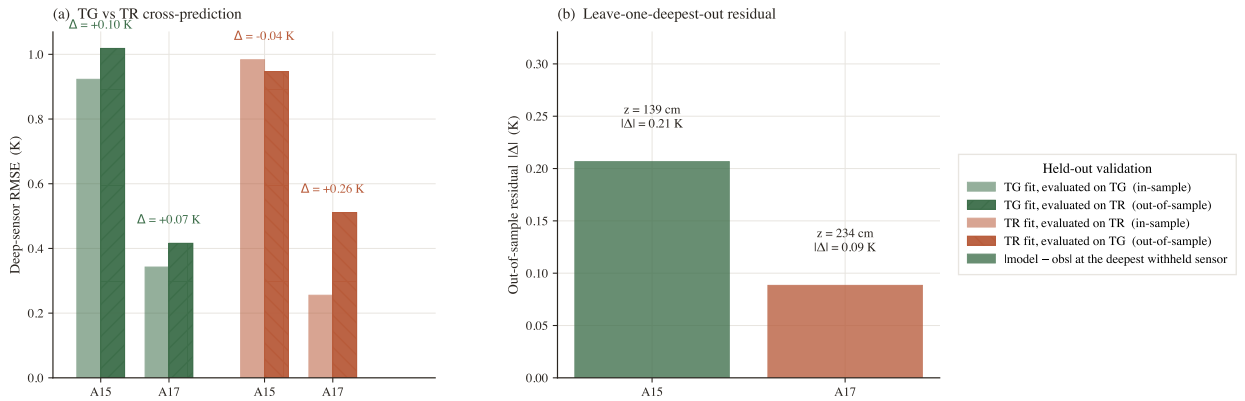


Figure S2. Held-out validation. (a) TG $\leftrightarrow$ TR cross-prediction RMSE (in-sample vs. out-of-sample) at both sites. (b) Absolute residual at the withheld deepest sensor. Generated by `scripts/pipeline/retrieve_kd.py`.

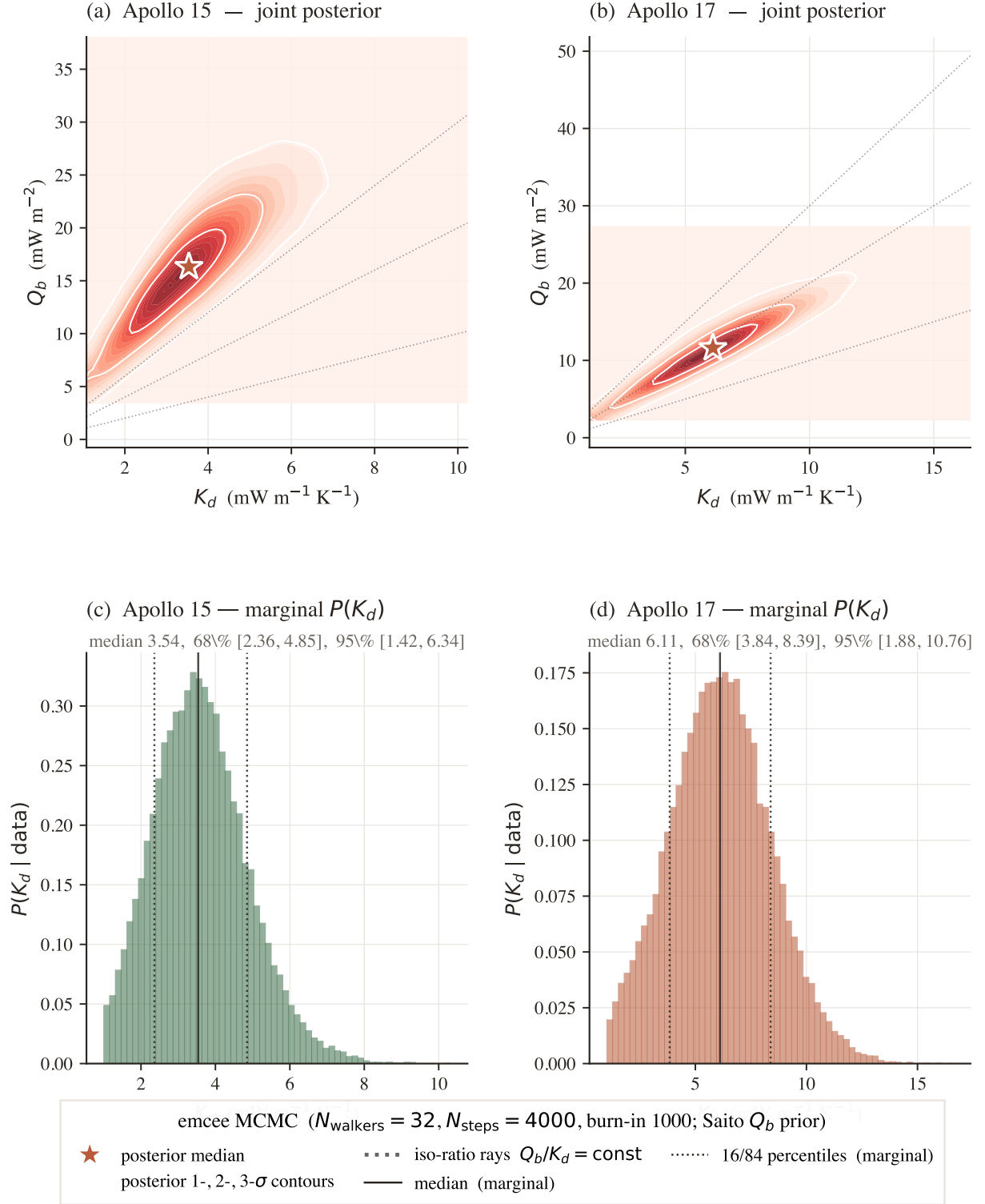


Figure S3. Joint (K_d, Q_b) posteriors at the two sites under the published basal-flux priors. The diagonal elongation is the $|\partial_z T| = Q_b/K_d$ degeneracy ridge. Generated by `scripts/pipeline/bayesian_crosscheck.py`.

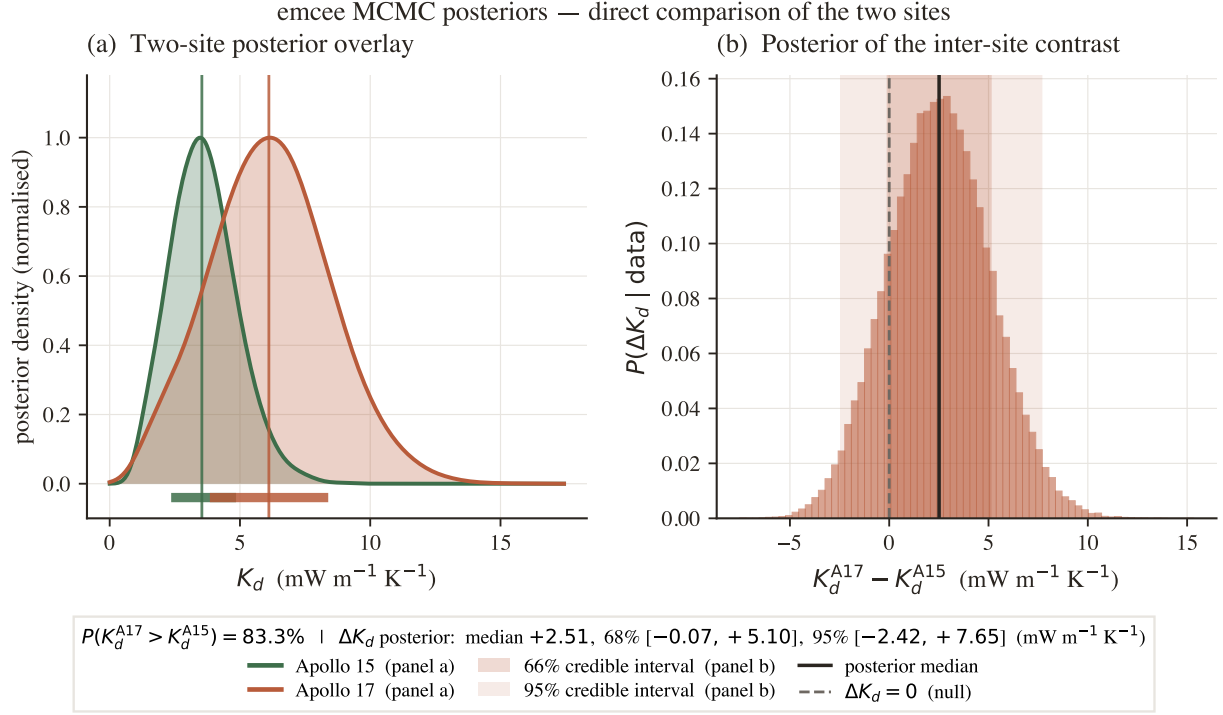


Figure S4. Marginal K_d posteriors compared with the bootstrap distributions of the main text (Fig. 4). Generated by scripts/pipeline/bayesian_crosscheck.py.

References

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